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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

02:19

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking** : Stack the two matrices horizontally (side by side).
- **Vertical Stacking** : Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

```
import numpy as np

# Input matrices
print("Enter Array1:")
arr1 = np.array([list(map(int, input().split())) for i in range(3)])

print("Enter Array2:")
arr2 = np.array([list(map(int, input().split())) for i in range(3)])

# Perform horizontal stacking (hstack)
print("Horizontal Stack:")
print(np.hstack((arr1, arr2)))

# Perform vertical stacking (vstack)
print("Vertical Stack:")
```

```
print(np.vstack((arr1, arr2)))
```

stacking.py

1 import numpy as np
2

Average time
0.030 s
30.00 ms

Maximum time
0.049 s
49.00 ms

✓ 2 out of 2 shown test case(s) passed
✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 26 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack:	Horizontal Stack:
[[1 2 3 4 5 6]]	[[1 2 3 4 5 6]]
[-4 -5 -6 -7 -8 -9]	[-4 -5 -6 -7 -8 -9]
[-7 -8 -9 10 11 12]	[-7 -8 -9 10 11 12]
Vertical Stack:	Vertical Stack:
[[1 2 3]]	[[1 2 3]]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[-4 -5 -6]	[-4 -5 -6]
[-7 -8 -9]	[-7 -8 -9]
[-10 -11 -12]	[-10 -11 -12]

✓ Test case 2 23 ms

Terminal

Test cases

< Prev

Reset

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